

YUETAO CHEN

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Education

Institute of Computing Technology, Chinese Academy of Sciences

September 2021 – June 2024

Master of Science in Computer Science

- GPA: 3.79 / 4.00

Beijing Jiaotong University

September 2017 – June 2021

Bachelor of Engineering in Computer Science

- GPA: 3.93 / 4.00 **Ranking:** 1 / 54

Lancaster University

September 2017 – June 2021

Bachelor of Science in Computer Science

- GPA: 3.81 / 4.00 **First class honours**

- 1 year at Bailrigg campus, UK; 3 years at Lancaster University College at Beijing Jiaotong University, China

Experience

Research Intern

March 2023 – Present

Microsoft Research

Beijing, China

Open Source Contributor

June 2022 – September 2022

OpenSUSE

Online

Back-End Development Intern

November 2017 – September 2018

Information Center, Beijing Jiaotong University

Beijing, China

Publications

* indicates the corresponding author.

- [1] **Yuetao Chen**, Kun Li*, Yuhao Wang, Donglin Bai, Lei Wang, Lingxiao Ma, Liang Yuan, Yunquan Zhang, Ting Cao, Mao Yang, ConvStencil: Transform Stencil Computation to Matrix Multiplication on Tensor Cores, *Submitted to PPOPP 2024* (under review)
- [2] Kun Li, Zhichun Li, **Yuetao Chen**, Zixuan Wang, Yiwei Zhang, Liang Yuan, Haipeng Jia, Yunquan Zhang, Ting Cao*, Mao Yan, Tetris: Exploiting More Parallelisms for Stencil Computation on Heterogeneous Architecture via Tetrominoes Tessellation, *Submitted to HPCA 2024* (under review) [old version link]
- [3] Yiwei Zhang, Kun Li*, Liang Yuan, **Yuetao Chen**, Haipeng Jia, Yunquan Zhang, Ting Cao, Jigsaw: Toward Conflict-free Vectorized Stencil Computation by Tessellating Swizzled Registers, *Submitted to PPOPP 2024* (under review)
- [4] Lei Liu*, Xinglei Dou, **Yuetao Chen**, Intelligent Resource Scheduling for Co-located Latency-critical Services: A Multi-Model Collaborative Learning Approach, *FAST 2023* [link]
- [5] **Yuetao Chen**, Keni Qiu, Li Chen, Haipeng Jia, Yunquan Zhang, Limin Xiao, Lei Liu*, Smart Scheduler: an Adaptive NVM-Aware Thread Scheduling Approach on NUMA Systems, *CCF Transactions on High Performance Computing* [link]

Research Projects

Transform Stencil Computation to Matrix Multiplication on Tensor Cores [1]

February 2023 – Present

Advisor: Dr. Kun Li | Microsoft Research

Beijing, China

- We proposed ConvStencil, a novel stencil computing system designed to efficiently transform stencil computation to matrix multiplication on Tensor Cores.
- Inspired by GEMM-based convolution and im2col, proposed *stencil2row* layout transformation which significantly saves memory footprint by over 70%.
- Proposed *dual tessellation* algorithm which increases Tensor Core utilization from 12.5% to 87.5%.
- Proposed *dirty bits padding* to reduce bank conflicts and branch diverges, improving the performance by 21.4%.

Mobile Platform Resource Scheduling for Optimizing UI Smoothness and Power

October 2022 – Present

Advisor: Prof. Lei Liu | Institute of Computing Technology, Chinese Academy of Sciences

Beijing, China

- It is called MobiRL, a reinforcement learning-based resource scheduler for mobile systems.
- Implemented a resource scheduling framework for Android devices using the Deep Deterministic Policy Gradient (DDPG) model to optimize performance.
- Designed a piecewise reward function that optimizes energy consumption while limiting frame loss rate, ensuring both efficient use of resources and a high-quality user experience.
- Evaluated by the OPPO mobile system development team, our MobiRL outperforms the current scheduler by reducing 38.1% frame loss rate and 3.1% power consumption.

Stencil Computing on Heterogeneous Platform [2][3]

October 2022 – December 2022

Advisor: Dr. Kun Li | Microsoft Research

Beijing, China

- It is the first system for high-performance stencil computation on heterogeneous CPU+GPU with novel optimizations on both CPU and GPU.
- Implemented a new CPU SIMD register level algorithm for stencil computing, resulting in significant performance improvements of 6.3% - 19.9% on different stencil kernel shapes.
- Referred to Tessellate Tiling (SC '17) and implemented a tiling algorithm, allowing vectorization and tiling algorithms to work together efficiently, handle boundary cases effectively, and reduce mutual interference between them.
- Designed a data exchange method between CPU and GPU for stencil computation which covers the communication latency by computation.

Intelligent Resource Scheduling for Co-located Services [4]

June 2021 – September 2022

Advisor: Prof. Lei Liu | Institute of Computing Technology, Chinese Academy of Sciences

Beijing, China

- It is called OSML which employs multiple ML models to work collaboratively to predict QoS variations, shepherd the scheduling, and recover from QoS violations in complicated co-location services.
- Divided the DQN model into two parts to prevent incorrect scheduling decisions and ensure the QoS.
- Designed a resource sharing policy, such as shared cacheline, to optimize resource utilization across co-located services.
- Tuned PARTIES (ASPLOS '19) and CLITE (HPCA '20) for the evaluation and collected and annotated program resource allocation data.

NVM-aware thread scheduling approach on NUMA systems [5]

September 2021 – May 2022

Advisor: Prof. Lei Liu | Institute of Computing Technology, Chinese Academy of Sciences

Beijing, China

- It is an NVM-aware thread scheduling approach for NUMA systems with hybrid memory, utilizing the LinUCB algorithm to guide scheduling decisions and reduce benchmark execution time by up to 59.9%.
- Found the significant difference in local and remote access bandwidth between NVM and DRAM led to performance loss for existing schedulers.

Fast Fourier Transforms Library

October 2020 – December 2020

Advisor: Prof. Haipeng Jia | Institute of Computing Technology, Chinese Academy of Sciences

Beijing, China

- Discovered the changing pattern of FFT algorithm with base 2.
- Automatically generated C language codes for FFT algorithm with base 2.

Development Experience

Dynamic Detection of Error Conditions from openQA Test Results

June 2022 – September 2022

OpenSUSE [Issue link] [PR link]

Online

- Communicated with the community and maintainers to refine what they want and evaluate how I did at the time.
- Added new functionality to openQA, enabling the system to search for similar failed test cases.
- Implemented visualization features in openQA to display failed tests and provide an intuitive understanding of common failures and their relationships, enabling more efficient and effective debugging.

WeCom Development

November 2017 – September 2018

Information Center, Beijing Jiaotong University

Beijing, China

- Leveraged Python, Django, and MySQL to implement functionalities in WeCom such as lost and found.

Teaching Assistant and Extracurricular Activity

Teaching Assistant of Linear Algebra

March 2020 – July 2020

Beijing Jiaotong University

- Graded homework and answered students' questions.

Deputy Director of the Academic Department of the Student Union

September 2022 – September 2023

Institute of Computing Technology, Chinese Academy of Sciences

- Invited professors to give talks and held graduate academic forums.

Technical Skills

Languages: C, Python, CUDA, C++

Tools and Framework: Linux, Git, NSight, CPU intrinsics, Intel CAT, cuDNN, Pytorch/libtorch, Django

Awards

E Fund Management Co. Scholarship (12/420)

Institute of Computing Technology, Chinese Academy of Sciences 2023

Postgraduate Academic Scholarship

University of Chinese Academy of Sciences 2021, 2022

Excellent Bachelor Thesis

Beijing Jiaotong University 2021

Second Class Academic scholarship

Beijing Jiaotong University 2020

First Class Academic scholarship (1/54)

Beijing Jiaotong University 2018, 2019